

WASP-43b

R-1271

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190331 · created 2026-07-31T13:55:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458573.703 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.014
Transit depth (Rp/Rs)^2	2.72 +/- 0.45 [%]
Orbital Inclination (inc)	82.69 +/- 1.13
Airmass coefficient 1 (a1)	1.099 +/- 0.011
Airmass coefficient 2 (a2)	-0.065 +/- 0.029
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	None
Optimal Aperture	5.1
Optimal Annulus	11.53
Transit Duration (day)	0.053 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.014 vs 0.1594 (deviation 0.37σ)
Duration fit vs published	0.053 d vs 0.0512 d (deviation 0.5σ)
Mid-time O–C	1.2 min
Depth SNR	10.8
Residual scatter	1.0 %

Light curve

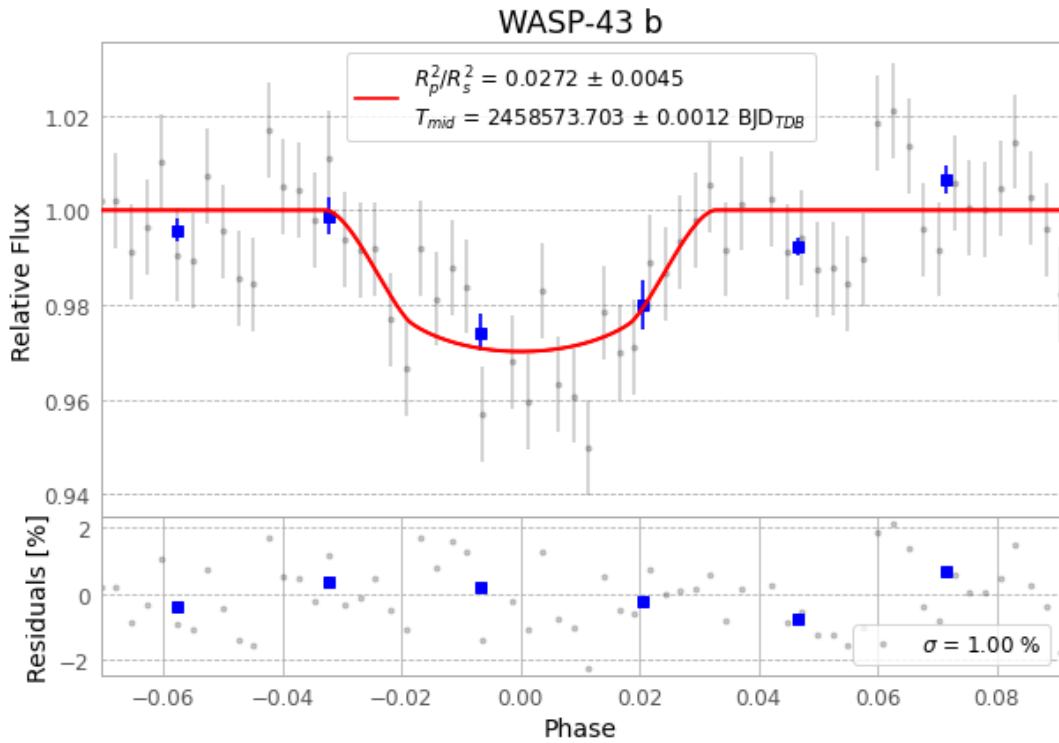


Figure 1 — FinalLightCurve_WASP-43 b_2019-03-30.png (SHA-256 6f60f85fa98da1cd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 142], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 38aff589b5af0f0f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43190331032112.FITS → WASP-43190331063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190331.log (9ff9a91ab1ce...), FinalLightCurve_WASP-43 b_2019-03-30.png (6f60f85fa98d...), FinalLightCurve_WASP-43 b_2019-03-30.csv (9e1c4f03fe56...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 13:55Z.